

Geometry of Factuality in LLM Internal Representations

Shay Tanne Lior Krinberg

Reichman University

{shay.tanne, lior.krinberg}@post.runi.ac.il

Submitted as final project report for the NLP course, Reichman University, 2026

1 Introduction

The rapid spread of misinformation in various information ecosystems poses severe societal threats [Lazer et al., 2018, Vosoughi et al., 2018]. Generative AI further amplifies the velocity and persuasiveness of deceptive claims [Spitale et al., 2023], making scalable information verification increasingly valuable. Conventional fact-checking relies on compute-heavy, latency-intensive external retrieval pipelines [Thorne et al., 2018, Zeng et al., 2021]. Mechanistic interpretability offers a compelling alternative: evaluating whether internal LLM activation states can serve as a fast, retrieval-free proxy to verify factuality [Azaria and Mitchell, 2023, Burns et al., 2023].

The feasibility of representation-based fact-checking depends on how truth is geometrically structured in LLM latent spaces. Earlier studies proposed an emergent, cross-domain, globally linear "geometry of truth" [Marks and Tegmark, 2024, Zou et al., 2023], suggesting a single direction separates true from false claims universally. In contrast, recent work demonstrates firm task-level orthogonality [Azizian et al., 2025]: probes often capture dataset-specific stylistic features rather than generalized truthfulness. Reconciling these views, the *Truthfulness Spectrum Hypothesis* [Ying et al., 2026] posits that model representations contain a spectrum ranging from domain-general to genre-specific directions.

However, prior work relies on synthetic text, ignores genre confounds, or considers only static activations. In this work, we systematically evaluate probing across real-world discourse fields: social rumors, biomedical claims, political debates, Wikipedia assertions, and web journalism. We utilize concept erasure to neutralize domain confounds, and evaluate geometries from static 1D directions, through k -dimensional subspaces, to dynamic representation evolution through the network.

Our experiments yield three primary contributions:

1. **Confound-Driven Transfer Collapse:** while linear probes decode in-domain factuality (65.6%–84.4%), neutralizing domain identity via INLP causes static domain transfer to collapse below majority baseline (54.0%), demonstrating that naive static transfer is primarily driven by shared stylistic confounds.
2. **Local 1D Geometry & Shared SVD Core:** in-domain truth is largely 1-dimensional (erasing $k = 1$ removes > 70 –80% of signal). Global SVD reveals a shared principal core direction (accounting for $> 48\%$ of the shared variance across layers 15–33 via singular value σ_1) translated by orthogonal domain offsets.
3. **Lie Detection in Language-Rich Discourse using Dynamic Trajectories:** layer-to-layer displacements filter out static domain offsets and isolate credibility updates. This rescues zero-shot deception detection in rhetoric-rich political discourse (LIAR: $\Delta = +17.9\%$), implying that activation **dynamics** capture signatures of real-world deception, while sterile entity-lookup benchmarks (FEVER) remain bound to the model’s parametric memory.

2 Methodology

Our investigation centers on *linear probing*—an interpretability technique that trains lightweight linear classifiers on top of frozen internal activations to test whether semantic properties like factual veracity are linearly accessible within the model’s latent representation space [Azaria and Mitchell, 2023, Marks and Tegmark, 2024].

2.1 Discourse Domains & Data Preprocessing

We evaluate binary (T/F) factuality probing on frozen internal model activations across five real-world natural discourse domains: **PHEME** (social rumors; Zubiaga et al. 2016), **SciFact** (biomedical science; Wadden et al. 2020), **LIAR** (political speeches; Wang 2017), **FEVER** (Wikipedia entity claims; Thorne et al. 2018), and **MultiFC** (open-web journalism; Augenstein et al. 2019). To prevent sample-size asymmetry and length biases from distorting computations, we applied two key controls:

1. **Equal Dataset Sizes (sample parity)**: every dataset was standardized to $N = 638$ claims, partitioned into 70% train, 15% validation, and 15% test splits, preventing size imbalances.
2. **Length Distribution Matching**: claims grouped into word-length bins and subsampled s.t. all share a matching length distribution, preventing length heuristics from affecting training.

2.2 Model Architecture & Extraction Protocol

We probe `google/gemma-4-E2B` (36 layers, hidden state dimension $d = 4224$, alternating sliding-window and global attention). For each claim x , hidden state activations $\mathbf{h}_l \in \mathbb{R}^d$ are extracted at the **final prompt token** across 12 uniformly-spaced depths: $L = \{3, 6, \dots, 33, 35\}$. We compare raw text claims (x) against a QA-formatted prompt: "Q: [x] True or False? A:".

2.3 Confound Removal (Domain Neutralization)

To verify whether cross-domain transfer reflects genuine factuality rather than shared writing styles or vocabulary artifacts, we neutralize domain-identifying signals in the internal activations. We employ two linear concept erasure techniques:

- **Iterative Nullspace Projection (INLP)** [Ravfogel et al., 2020]: iteratively identifies and removes latent space directions that encode domain identity. In each round, a linear classifier is trained to predict which domain a claim belongs to, and activations are projected onto the subspace orthogonal to the classifier’s weights vector (nullspace of weight matrix W when rounds are stacked). This process is repeated until a domain classifier can no longer distinguish domains better than random chance ($\leq 50.5\%$ accuracy, requiring 12 iterations in this project).
- **One-Shot Mean Difference Removal**: following LEACE [Belrose et al., 2023], calculates the difference between the average vector representations of two domains (the centroid shift) and removes that single direction in a single projection step.

To prevent data leakage, all projection operators and feature scalers are computed strictly on the training and validation sets before being applied to held-out test data.

2.4 Dynamic Trajectory Probing

While standard static probing evaluates a single activation snapshot at an isolated layer l ($\mathbf{h}_l$), dynamic trajectory probing tracks how activation vectors evolve as they travel through transformer layers. Background stylistic and domain features remain largely constant across adjacent layers, and

so computing the change between layers can effectively cancel out static domain offsets, isolating the internal reasoning updates that resolve factuality.

We evaluate three trajectory formulations:

- **Step Displacement** ($\Delta \mathbf{h}_{l \rightarrow l+k} = \mathbf{h}_{l+k} - \mathbf{h}_l$): measures the immediate vector change between two layers.
- **Cumulative Drift** ($\mathbf{d}_l = \mathbf{h}_l - \mathbf{h}_3$): measures total displacement from the initial input layer ($L3$) to layer l .
- **Depth Bands**: concatenates trajectory updates across depth segments (Early: $L3$ – 12 , Mid: $L15$ – 24 , Late: $L27$ – 35).

2.5 Implementation, Platform & Challenges

The pipeline is implemented in the `factgeom` package (using PyTorch, `transformers`, `scikit-learn`). Activations were extracted on Google Colab (T4 GPU runtime, ~ 1.5 h for 3,190 claims); linear probe sweeps (135 runs) were executed locally on a 24-thread CPU (~ 1.0 h).

Key technical hurdles included matrix-free projection computation (to prevent memory overflow with $d = 4224$) and zero-leakage cross-validation hygiene.

3 Experimental results

Experiment Settings & Metrics: probes use L_2 regularized logistic regression ($C = 1.0$, verified via $C \in [10^{-3}, 10^2]$ sweep) evaluated on held-out test splits ($N_{\text{test}} = 96$, seed = 42). We report Accuracy (Acc), 95% bootstrap CIs (1,000 resamples), Macro-F1 ($= \frac{F1_{\text{true}} + F1_{\text{false}}}{2}$), AUROC, gain over maj. baseline ($\Delta Acc = Acc - \text{Baseline}_{\text{maj}}$), and binomial p -values against the maj. baseline.

3.1 Exp 0: In-Domain Learnability & Emergence Dynamics

Main Idea: before evaluating cross-domain generalization, we establish a baseline to verify whether linear probes can reliably decode claim truthfulness within each individual domain and identify the network layers where factual signals peak.

Results: as shown in Table 1, linear probes effectively separate truth in-domain across all five domains, substantially beating majority baselines (see the ΔAcc column in Table 1). The QA prompt format boosts accuracy by 2.1%–3.2% and ~ 0.035 AUROC across all domains, possibly by pointing attention to factuality earlier in the forward pass.

Table 1: In-Domain Factuality Probing (Exp 0). Held-out test split ($N = 96$) with 95% CIs.

Domain	Majority	Raw Acc	QA Acc	Peak Layer	ΔAcc	F1	AUROC	95% CI
PHEME (Rumors)	60.4%	80.2%	82.3%	$L35$ (Late)	+21.9%	0.850	0.873	[74.0%, 89.6%]
FEVER (Wikipedia)	70.8%	82.3%	84.4%	$L33$ (Late)	+13.6%	0.895	0.878	[76.0%, 91.7%]
LIAR (Politics)	53.1%	63.5%	65.6%	$L15$ (Mid)	+12.5%	0.629	0.721	[55.2%, 75.0%]
SciFact (Biomedical)	65.3%	70.5%	73.7%	$L18$ (Mid)	+8.4%	0.800	0.746	[64.2%, 82.1%]
MultiFC (Open-Web)	70.8%	75.0%	77.1%	$L12$ (Early-Mid)	+6.3%	0.542	0.723	[68.8%, 85.4%]

Tracking probe accuracy across network depth reveals two distinct emergence patterns:

- **Middle-Layer Peak (Complex Reasoning):** in domains requiring deeper semantic evaluation – MultiFC (peak at $L12$), LIAR ($L15$), and SciFact ($L18$) – factuality is most accessible in the middle-band layers. Accuracy then declines by 4.5%–12.6% in the final layers ($L35$) as representations shift toward formatting next-token generation.

- **Late-Layer Peak (Direct Facts & Stance)**: in contrast, straightforward Wikipedia assertions (FEVER, *L33*) and social media rumors (PHEME, *L35*) reach peak decodability in the deepest layers, where the model finalizes its surface "arguments".

3.2 Exp 1: Static Cross-Domain Transfer & Confound Collapse

Main Idea: We test whether a factuality probe trained on one domain transfers directly to another, and apply INLP domain neutralization to determine whether observed transfer stems from genuine generalized truth detection or shared stylistic confounds.

Results: Raw static representations appear to transfer across domains (mean 60.2%, Figure 1), e.g., SciFact $\rightarrow$ FEVER reaches 74.7%. However, when domain identity is removed via INLP ($T = 12$, dev accuracy $\leq 50.5\%$), **cross-domain transfer collapses to 54.0%** (Figure 1), falling below the 63.7% mean majority baseline. Pooled Leave-One-Domain-Out (LODO) probes under INLP fail similarly (SciFact: 49.5%, FEVER: 58.9%).

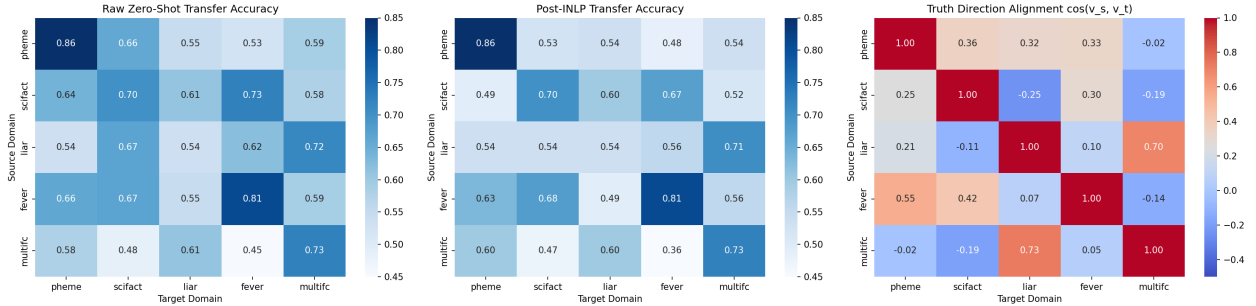


Figure 1: Pairwise Transfer Heatmaps (Exp 1)

To test whether domains share a single truth axis, we erased the truth vector learned on one domain and tested the effect on the remaining domains (Figure 2). If a universal truth direction existed, erasing it would harm truth detection across all domains. Instead, erasing a domain’s vector only degraded accuracy on that specific domain, having insignificant impact on the others ($\leq 3.2\%$ accuracy change). This suggests that static truth directions are domain-specific rather than universally aligned.

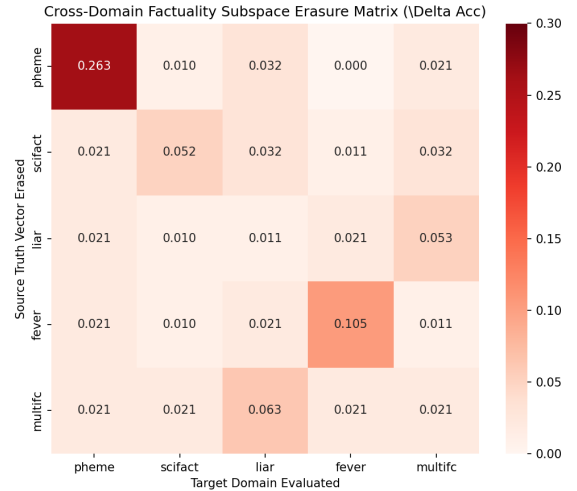


Figure 2: Subspace Erasure Overlap (Exp 1)

3.3 Exp 2: Subspace Dimensionality & SVD Analysis

Main Idea: to understand the geometry behind transfer failure, we investigate whether truth in each domain is encoded along a single line (via Rank- k erasure) and test whether domains share a common geometric core direction (via SVD analysis).

Results:

- **In-Domain Truth – 1-Dimensional:** erasing a single direction ($k = 1$) from activations causes in-domain accuracy to drop by 70%–80% effectively down to baseline (Figure 3a). This implies that within any individual domain, factuality is captured along a single 1D direction.
- **Shared Geometric Core:** when we combine the truth vectors from all five domains and analyze them using SVD, the primary singular direction (σ_1) accounts for 48%–56% **of the shared variance** across layers 15–33 (Figure 3b).

This shows that truth representations across domains are not completely unrelated – they share a common central orientation, but each domain is shifted by its own stylistic offset.

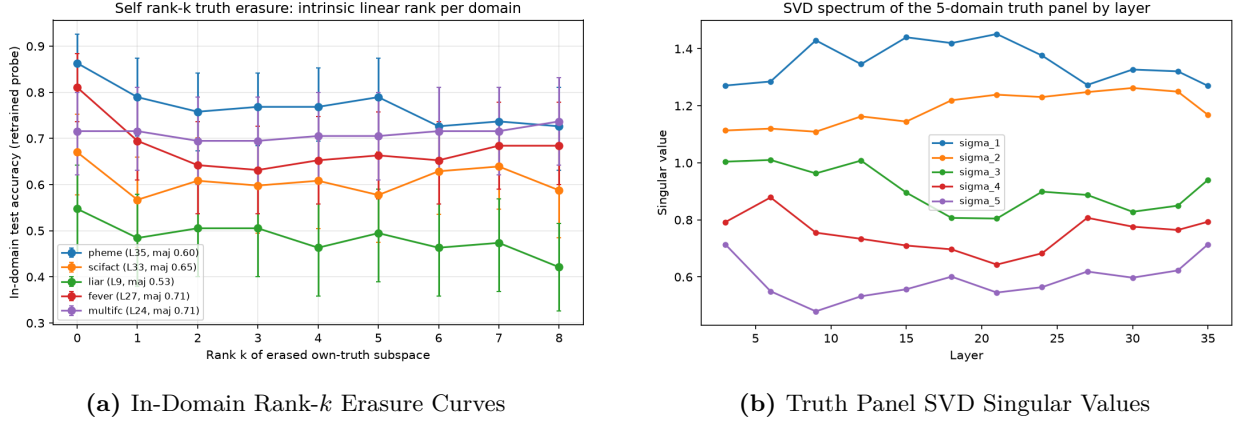


Figure 3: Subspace Geometry & SVD (Exp 2)

3.4 Exp 2A: Dynamic Trajectories & Discourse Rescue

Main Idea: because static activation snapshots mix together background writing style and factuality, we evaluate whether tracking layer movement ($\Delta \mathbf{h}$) can cancel out static domain offsets and "rescue" zero-shot transfer.

Results: Subtracting activations between layers ($\Delta \mathbf{h}_{15 \rightarrow 18}$) and measuring cumulative drift ($\mathbf{h}_{24} - \mathbf{h}_3$) removes the constant background style of each domain (Figure 4).

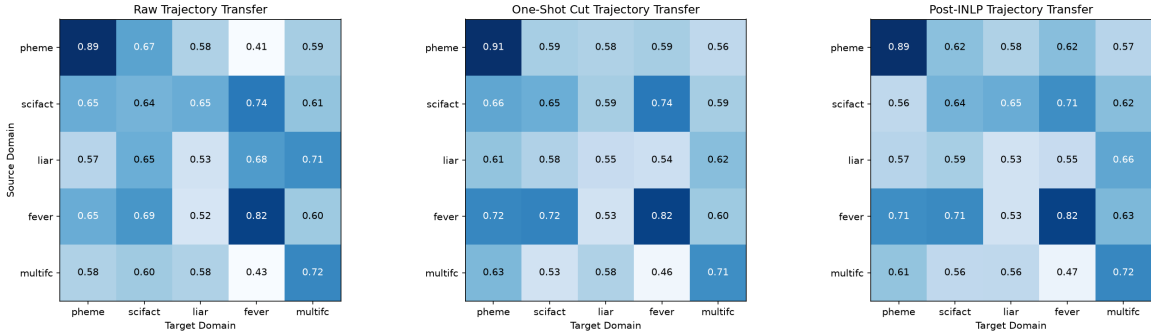


Figure 4: Pairwise Trajectory Transfer (Exp 2A)

Tracking trajectory updates across network depth shows that factuality resolution is concentrated in the Mid-Late layer bands (L_{12} – L_{27} , Figure 5). As detailed in Table 2:

- **Rescue on Politics (LIAR):** while static probes failed (54.7%), trajectory probes achieved 70.5% **accuracy**, $\Delta = +17.9\%$ post-INLP. Results with one-shot mean shift removal match INLP (70.5%), which supports that transfer failure was a simple positional offset between domain latent spaces.
- **Limits on Sterile Facts:** Wikipedia (FEVER: 64.2%) and web-journalism claims (MultiFC: 68.4%) remained near majority baselines.

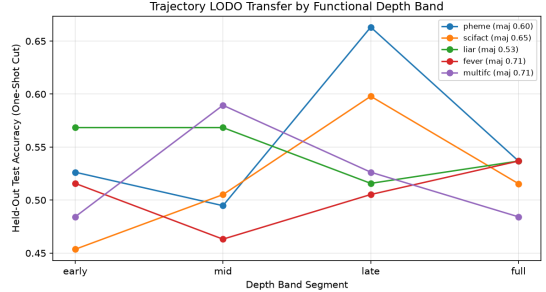


Figure 5: Trajectory Depth Bands (Exp 2A)

Pragmatic Discourse vs. Parametric Memory: The contrast between LIAR ($\Delta = +17.9\%$) and FEVER ($\Delta = -6.3\%$) reveals what trajectory probes actually decode. In language-rich discourse (LIAR), deception is accompanied by pragmatic and rhetorical markers (hyperbole, subjective framing, conviction shifts) that the model resolves dynamically across intermediate layers (L12–L18). Trajectory probing ($\Delta\mathbf{h}$) strips away background genre style and isolates this internal credibility assessment. In contrast, in sterile/declarative assertions (FEVER), true and false claims share identical grammar and neutral tone (e.g. “*Tom Hanks starred in Forrest Gump*” vs. “*Tom Hanks starred in The Matrix*”). Truthfulness leaves little rhetorical footprint in the language, and requires looking up specific entity-relation links stored in the model’s parametric weights (parametric memory). Because a low-dimensional probe direction cannot act as a universal database lookup, probes fail on dry entity facts while succeeding as lie detectors in rich discourse ecosystems.

Table 2: Held-Out LODO Generalization (Exp 1 vs. 2A)

Held-Out Domain	Majority	Static LODO	Raw Traj	1-Shot Traj	INLP Traj	Optimal Segment	p-value	Hypothesis Verdict
LIAR	52.6%	54.7%	70.5%	70.5%	70.5% [61.1%, 78.9%]	Cum. Drift (L3 → L12)	$p = 0.0002$	Pass (dynamic rescue +17.9%)
PHEME	60.0%	42.1%	60.0%	65.3%	63.2% [52.6%, 72.6%]	Cum. Drift (L3 → L15)	$p = 0.2810$	Marginal (+5.3% over maj)
SciFact	64.9%	50.5%	67.0%	61.9%	66.0% [55.7%, 75.3%]	Step Displ. (L27 → L30)	$p = 0.4250$	Marginal (+1.1% over maj)
MultiFC	70.5%	60.0%	71.6%	57.9%	68.4% [58.9%, 76.8%]	Step Displ. (L21 → L24)	$p = 0.7100$	Limited (domain local)
FEVER	70.5%	58.9%	60.0%	63.2%	64.2% [54.7%, 73.7%]	Cum. Drift (L3 → L24)	$p = 0.9200$	Limited (domain local)

4 Discussion

4.1 Reconciling Prior Theories

Our project explored the tension between universal linear truth [Marks and Tegmark, 2024] and task-specific truth [Azizian et al., 2025] within the *Truthfulness Spectrum Hypothesis* [Ying et al., 2026]:

- **Static** probes capture **domain-specific** features and fail transfer (collapsing to 54.0% under INLP).
- **SVD** analysis isolates the shared, **domain-general** core direction ($\sigma_1 > 48\%$).
- **Dynamic trajectory** probing ($\Delta\mathbf{h}$) isolates internal reasoning updates, acting as an effective zero-shot “lie detector” in language-rich discourse ecosystems (LIAR: 70.5%). In contrast, dry entity-level verification (FEVER) remains bound to parametric retrieval.

4.2 Reflection on the Scientific Process

Empirically contradicting our initial expectation of universal static transfer was a key project milestone. Rather than treating transfer failure as a negative outcome, removing confounds via INLP revealed that static representations are heavily tied to domain style. This motivated exploring deeper geometric and trajectory paths, confirming that reasoning peaks at middle layers (L12–L18)

and showing that representation evolution over the model ($\Delta\mathbf{h}$) decouples factual reasoning from writing style in natural human discourse.

4.3 Methodological Defenses & Statistical Rigor

Our claims rely on three assumptions/controls:

1. **Sample parity** ($N = 638$) across all five domains to prevent dataset size imbalances from distorting training and computations.
2. **Majority baseline evaluation** (ΔAcc) and 95% CIs ($N = 96$) to avoid misleading accuracy claims on skewed datasets.
3. **Regularization robustness** ($C \in [10^{-3}, 10^2]$, stable within $\pm 1\%$ – 2%), confirming results are not artifacts of specific hyperparameter selection.

4.4 Limitations & Future Work

While trajectory probing rescued transfer in language-rich political discourse (LIAR), encyclopedic fact-checking (FEVER) remained bound to domain characteristics. Furthermore, this study focused on a single model (google/gemma-4-E2B). Future work should test other model families (LLaMA-3, Mistral, Qwen), as well as applications like mitigating hallucinations by activation steering.

5 Conclusion

This work demonstrates that the geometry of factuality in LLMs is neither purely universal nor purely domain-specific, but occupies a structured spectrum. Static truth directions are domain-private and orthogonal, yet share a principal SVD core. Dynamic trajectory probing isolates internal credibility reasoning from static genre style, enabling zero-shot deception detection in rhetoric-rich discourse while establishing that encyclopedic entity verification remains bound to parametric memory. These findings chart a principled operational boundary for activation-based fact-checking and motivate future work on multi-model validation and activation steering for hallucination mitigation.

6 Code

- **GitHub Repository:** <https://github.com/lior-krinberg-msc/factgeom>
 - **Colab Replication Notebook:** [notebooks/00_run_official_suite.ipynb](https://colab.research.google.com/notebooks/00_run_official_suite.ipynb) (135 verified runs, `seed=42`).
- **Pre-Extracted Activations & Artifacts** (Google Drive): <https://drive.google.com/drive/folders/14KjmAmy1JsZvt0m6RFYpR63KFDH0ndz?usp=sharing>.

References

- Isabelle Augenstein, Christina Lioma, Doryan Wang, Lucas Chaves Lima, Casper Hansen, Christian Hansen, and Jakob Grue Simonsen. MultiFC: A real-world multi-domain dataset for evidence-based fact checking of claims. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 4685–4697, 2019. URL <https://aclanthology.org/D19-1475/>.
- Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it’s lying. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 967–976, 2023. URL <https://arxiv.org/abs/2304.13734>.

- Waiss Azizian, Lucas Caccia, Pau Rodriguez, and Simon Lacoste-Julien. The geometries of truth are orthogonal across tasks. *arXiv preprint arXiv:2506.08572*, 2025. URL <https://arxiv.org/abs/2506.08572>.
- Nora Belrose, David Furman, Logan Smith, Danny Halawi, Igor Kunnamkumarath, Rohin Shah, and Samuel Bowman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 50840–50867, 2023. URL <https://arxiv.org/abs/2306.03819>.
- Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://arxiv.org/abs/2212.03827>.
- David M. J. Lazer, Matthew A. Baum, Yochai Benkler, Adam J. Berinsky, Kelly M. Greenhill, Filippo Menczer, Miriam J. Metzger, Brendan Nyhan, Gordon Pennycook, David Rothschild, et al. The science of fake news. *Science*, 359(6380):1094–1096, 2018. doi: 10.1126/science.aao2998. URL <https://doi.org/10.1126/science.aao2998>.
- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *First Conference on Language Modeling (COLM)*, 2024. URL <https://arxiv.org/abs/2310.06824>.
- Shauli Ravfogel, Yanai Elazar, Hila Geva, Michael Cox, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 7256–7267, 2020. URL <https://arxiv.org/abs/2004.07667>.
- Giovanni Spitale, Nikola Biller-Andorno, and Federico Germani. AI model GPT-3 (dis)informs us better than humans. *Science Advances*, 9(26):eadh1850, 2023. doi: 10.1126/sciadv.adh1850. URL <https://doi.org/10.1126/sciadv.adh1850>.
- James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. FEVER: a large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 809–819, 2018. URL <https://arxiv.org/abs/1803.05355>.
- Soroush Vosoughi, Deb Roy, and Sinan Aral. The spread of true and false news online. *Science*, 359(6380):1146–1151, 2018. doi: 10.1126/science.aap9559. URL <https://doi.org/10.1126/science.aap9559>.
- David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hananeh Hajishirzi. Fact or fiction: Verifying scientific claims against medical literature. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3201–3211, 2020. URL <https://aclanthology.org/2020.emnlp-main.293/>.
- William Yang Wang. “liar, liar pants on fire”: A new benchmark dataset for fake news detection. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 422–426, 2017. URL <https://aclanthology.org/P17-2067/>.
- Zhuofan Josh Ying, Shauli Ravfogel, Nikolaus Kriegeskorte, and Peter Hase. The truthfulness spectrum hypothesis. *arXiv preprint arXiv:2602.20273*, 2026. URL <https://arxiv.org/abs/2602.20273>.
- Xia Zeng, Amani S. Abumansour, and Arkaitz Zubiaga. Automated fact-checking: A survey. *Language and Linguistics Compass*, 15(10):e12438, 2021. URL <https://arxiv.org/abs/2109.11427>.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xu Wang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023. URL <https://arxiv.org/abs/2310.01405>.
- Arkaitz Zubiaga, Maria Liakata, Rob Procter, Geraldine Wong Sak Hoi, and Peter Tolmie. Analysing how people orient to and spread rumours in social media by looking at conversational threads. *PLOS ONE*, 11(3):e0150989, 2016. doi: 10.1371/journal.pone.0150989. URL <https://doi.org/10.1371/journal.pone.0150989>.